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Perspectives for HET physics studies at a muon-hadron collider at CERN

Hamzeh Khanpour[a,*]
(on behalf of HELP4CERN Study Group)

[a] *AGH University of Krakow, Faculty of Physics and Applied Computer Science, Al. Mickiewicza 30, 30-055 Kraków, Poland.*

E-mail: hamzeh.khanpour@cern.ch

A muon-hadron collider at CERN would combine the energy reach of a TeV-scale muon beam with the proton beam of the High-Luminosity LHC, opening a programme of Higgs, electroweak and top (HET) studies. We discuss the HELP4CERN prospects for muon energies of 0.5, 1 and 2 TeV, corresponding to centre-of-mass energies of approximately 3.74, 5.3 and 7.5 TeV for 7 TeV protons. The accessible deep-inelastic-scattering (DIS) phase space and inclusive production rates provide the basis for the physics programme. Particular attention is given to a dedicated charged-current $h \to b\bar{b}$ study and an exploratory combined Higgs-coupling fit, distinguishing statistical rate precision from coupling constraints that depend on external information and fit assumptions. Electroweak and top measurements, photon-induced production, and muon-coupled new-physics channels offer complementary directions.

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*Speaker

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1. Collider benchmarks and HET production rates

The proposed LHμC would collide a positive-muon beam with the 7 TeV proton beam of the High-Luminosity LHC, extending the lepton-hadron programme beyond the electron-beam configurations of the LHeC [1–3]. This contribution focuses on the Higgs, electroweak and top (HET) programme, complemented by photon-induced reactions and selected new-physics opportunities [4, 5]. The inclusive production cross-sections for the Higgs and top-quark are reported in Table 1. The large charged-current (CC) and neutral-current (NC) Higgs samples and the increasing single-top and top-pair rates motivate detailed measurements.

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(\ell p \to \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \to \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \to \ell t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \to \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

Table 1: Benchmark inclusive production cross-sections for Higgs-boson, top-quark-pair, and single-top production at the LHeC and LHμC. In the LHeC cases an 80% electron beam polarisation is assumed [6].

The LHμC working points are summarized in Table 2. The first option employs a 500 GeV muon beam, giving a centre-of-mass energy of approximately 3.7 TeV and a projected integrated luminosity of 250 fb^{-1}. The nominal baseline uses a 1 TeV muon beam, corresponding to $\sqrt{s_{\mu p}} \simeq$ 5.3 TeV, with a benchmark integrated luminosity of 1000 fb^{-1}. The second option increases the muon-beam energy to 2 TeV, reaching $\sqrt{s_{\mu p}} \simeq 7.5$ TeV with the same benchmark integrated luminosity.

Configuration	E_μ	E_p	$\sqrt{s_{\mu p}}$	$\mathcal{L}_{\text{int}}$
First option	500 GeV	7 TeV	3.7 TeV	250 fb^{-1}
Nominal baseline	1 TeV	7 TeV	5.3 TeV	1000 fb^{-1}
Second option	2 TeV	7 TeV	7.5 TeV	1000 fb^{-1}

Table 2: Benchmark LHμC working points based on the HELP4CERN studies [5]. The integrated luminosities are projected benchmark samples.

These working points illustrate the potential of the LHμC physics programme. Already with a 500 GeV muon beam, collisions with 7 TeV protons enter the multi-TeV centre-of-mass regime, providing a basis for competitive studies of Higgs production, electroweak interactions, top-quark processes, photon-induced reactions, and representative scenarios beyond the Standard Model. The nominal baseline combines a higher collision energy with a larger benchmark data sample, supporting more demanding precision measurements and searches. The 2 TeV muon-beam option further extends the accessible momentum transfer and final-state invariant masses. Together, these configurations provide a framework for examining how the physics opportunities develop with collision energy and integrated luminosity.

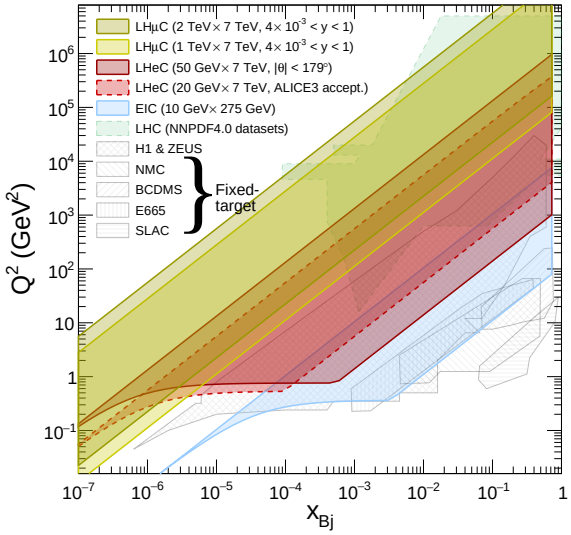
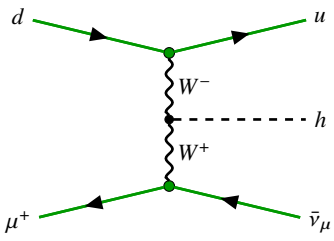


Figure 1: Accessible (x_{Bj}, Q^2) plane from HELP4CERN [5], compared with existing datasets and planned EIC kinematics. The LHμC regions use $4 \times 10^{-3} < y < 1$. The LHeC curves retain their plotted 20/50 GeV energies and acceptance conventions.

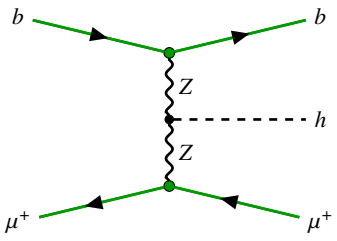
Figure 1 shows the accessible (x_{Bj}, Q^2) region. The enlarged high-Q^2 domain supplies both the hard-scattering phase space for HET production and sensitivity to electroweak effects in deep-inelastic scattering (DIS); it complements the broader QCD programme rather than replacing it.

2. Higgs production and precision prospects

Charged-current (CC) Higgs production proceeds through W^+W^- fusion. The channel $\mu^+ p \to j \bar{\nu}_\mu h$ is characterized by an outgoing quark jet, typically in the forward region, and missing momentum carried by the antineutrino. A representative partonic contribution is shown in Fig. 2(a). Neutral-current (NC) production, $\mu^+ p \to \mu^+ j h$, proceeds through ZZ fusion and retains the scattered charged lepton, providing a complementary experimental signature; an example is shown in Fig. 2(b).



(a) Charged-current production



(b) Neutral-current production

Figure 2: Representative tree-level diagrams for Higgs production in positive-muon–proton collisions: (a) the charged-current subprocess $\mu^+ d \to \bar{\nu}_\mu u h$ through W^+W^- fusion; (b) the neutral-current subprocess $\mu^+ b \to \mu^+ b h$ through ZZ fusion. The outgoing quark contributes to the jet signature.

The rates in Table 1, however, establish the production potential. Extracting Higgs couplings additionally requires decay information, background control, and explicit assumptions about the Higgs width.

2.1 A dedicated $h \to b\bar{b}$ study

Following the approach of Ref. [7], we study $\mu^+ p \to j\bar{\nu}_\mu h$, $h \to b\bar{b}$, against CC multijet production and weak-boson-fusion $Z \to b\bar{b}$. We uses an LHeC-like fast detector simulation with Delphes [8], a stated 75% b-tagging factor, exactly two selected b-jets, and $|m_{bb} - m_h| < 25$ GeV. The quoted rate precision includes statistical errors only:

$$\frac{\Delta\sigma}{\sigma} = \frac{\sqrt{N_s + N_b}}{N_s}, \tag{1}$$

where N_s and N_b are the selected signal and background counts. The source reports 1.19% and 0.69% for its 500 GeV and 1 TeV muon studies, respectively. The conversion to a bottom-Yukawa constraint uses external information on $R = g^2_{hWW}/\Gamma_h$, where Γ_h is the total Higgs width. Since the measured production-times-decay rate scales as $R\, g^2_{hbb}$, the source combines independent relative uncertainties in quadrature:

$$\frac{\Delta g_{hbb}}{g_{hbb}} = \frac{1}{2}\sqrt{\left(\frac{\Delta\sigma}{\sigma}\right)^2 + \left(\frac{\Delta R}{R}\right)^2}. \tag{2}$$

With the external $\Delta R/R = 0.985\%$ input following the FCC-ee benchmark treatment of Ref. [7], we obtain coupling precision of 0.77% and 0.60%.

2.2 Exploratory combined coupling fit

We perform a complementary combined Higgs-coupling fit in the κ-3 framework [5, 9, 10]. For this exploratory study, the statistical components of the LHeC signal-strength uncertainties are rescaled using the Higgs-production rates, while retaining the source assumptions about backgrounds, acceptance, efficiency, and systematic factors. The results are expressed in terms of fermion, vector-boson, and effective loop-induced coupling modifiers, together with constraints on invisible and undetected Higgs decays.

Figure 3 summarizes the exploratory κ-3 fit [5, 9], performed with SMEFiT [10] by rescaling the LHeC signal-strength uncertainties while retaining the source assumptions about backgrounds, acceptance, efficiency, and systematic factors. The LHeC+LHμC combinations are compared with the LHeC reference and HL-LHC baselines [11]; "HL-LHC improved" incorporates reduced PDF and α_s uncertainties [3]. The combined κ_W uncertainties reach 0.28%, 0.19%, and 0.16% for the 500 GeV, 1 TeV, and 2 TeV stages, respectively. At 2 TeV, the reported uncertainties on κ_b, κ_c, κ_g, and κ_γ are 0.50%, 1.2%, 0.75%, and 1.1%, whereas the improvements in κ_t, κ_μ, and $\kappa_{Z\gamma}$ are smaller. Since these measurements do not independently determine the Higgs width, the fit imposes $\kappa_V(\kappa_W, \kappa_Z) \leq 1$, with vector uncertainties defined as $1 - \kappa_V(68\%)$. These are model-dependent combined projections, distinct from the dedicated $h \to b\bar{b}$ extraction and from standalone LHμC sensitivities.

Taken together, these projections illustrate the complementary information provided by the different Higgs channels. The higher-energy combinations improve several fermion, vector, and

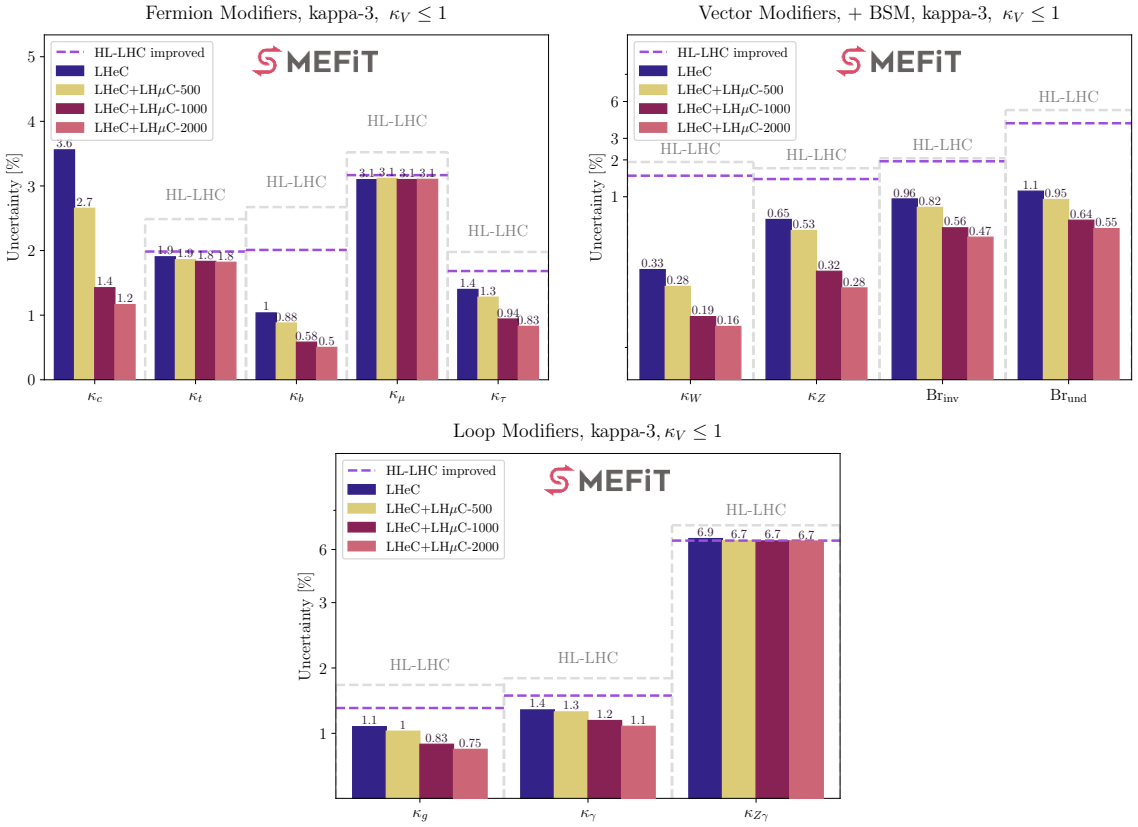


Figure 3: Exploratory κ-3 fit projections from HELP4CERN [5]: fermion modifiers (upper left), vector modifiers and invisible/undetected branching fractions (upper right), and effective loop-induced modifiers (lower). Results are shown for the LHeC and combined LHeC+LHμC stages, with HL-LHC baselines. Coupling uncertainties and branching-fraction constraints are expressed in percent; the fit assumes $\kappa_V \leq 1$.

loop-induced coupling constraints, although the size of the improvement varies substantially between observables. The results remain exploratory combined projections: they do not represent standalone LHμC measurements or independent detector-level sensitivity studies for every channel. Establishing a complete precision programme requires dedicated analyses of detector response, backgrounds, systematic uncertainties, and their correlations.

3. Electroweak and top physics

The extension of DIS to high momentum transfer provides a complementary setting for testing the electroweak interaction. NC and CC measurements probe different combinations of quark flavours and electroweak couplings, while their kinematic dependence can help separate changes in the hard interaction from proton-structure effects. Weak-mixing-angle studies and constraints on effective interactions are therefore motivated by the available phase space [2, 5].

In the top sector, CC single-top production gives access to the charged-current interaction involving the top quark and motivates studies of the Wtb vertex and CKM parameters. NC top-pair production offers a distinct final state and probes the top sector through complementary production

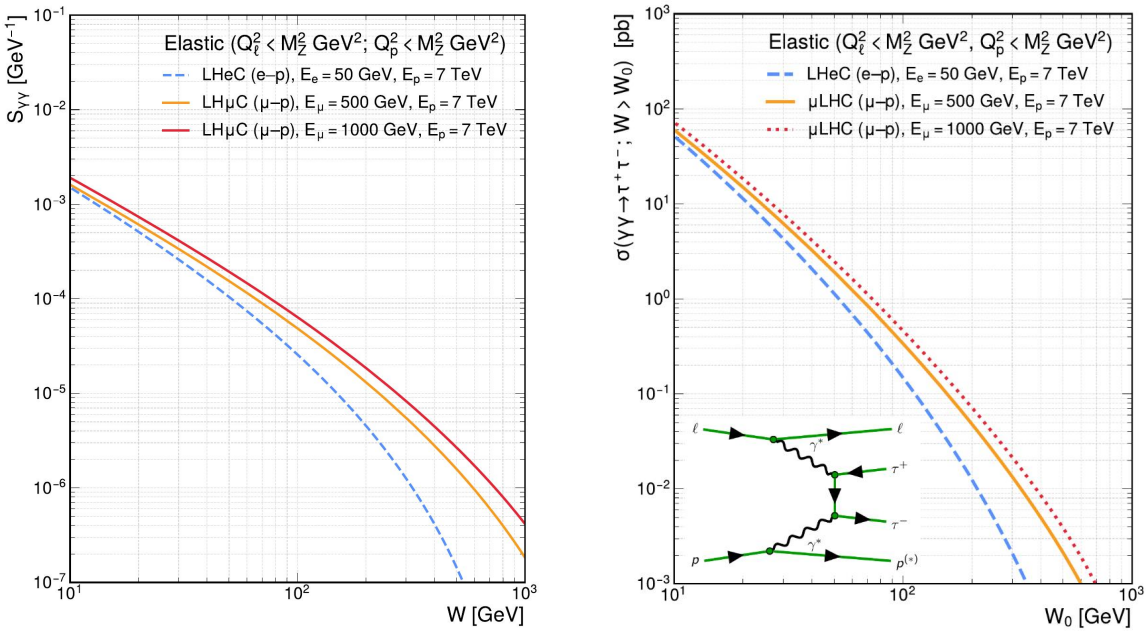


Figure 4: Elastic effective photon luminosity (left) and cumulative τ-pair cross-section $\sigma(W > W_0)$ (right). The curves compare 50 GeV electrons and 500/1000 GeV muons on 7 TeV protons.

and decay observables [5]. Table 1 displays the source-reported growth of both rates with muon energy.

Studies of anomalous couplings, flavour-changing neutral currents and rare top decays are possible necessarily extensions of this programme. They should be formulated through explicit observables and signal/background analyses rather than by translating inclusive rates directly into limits. The expected clean lepton-hadron environment is advantageous, and dedicated studies need for tagging, acceptance calibration and systematic control.

4. Photon-induced physics

Photon-induced reactions add a complementary production mechanism [5, 12, 13]. The equivalent-photon approximation (EPA) describes the photon fluxes through an effective photon-photon luminosity $S_{\gamma\gamma}(W)$. Here W is the photon-pair invariant mass, distinct from $\sqrt{s_{\mu p}}$. To calculate the lepton-proton level cross section, the $S_{\gamma\gamma}$ need to be convoluted with the relevant subprocess cross-section:

$$\sigma(\mu p \to \mu X p; W > W_0) = \int_{W_0}^{W_{\max}} dW\, S_{\gamma\gamma}(W)\, \hat{\sigma}(\gamma\gamma \to X; W). \tag{3}$$

Here $W_{\max}$ is the kinematic endpoint. The discussion here concerns elastic proton emission with the source photon-virtuality restrictions; it does not imply detector selection or proton-tagging efficiency. Figure 4 illustrates the photon luminosity and cumulative τ-pair production.

For $\tau^+\tau^-$ production above $W_0 = 10$ GeV, we calculate the cumulative cross-sections of 50.37, 58.56 and 70.3 pb for 50 GeV electrons, 500 GeV muons and 1 TeV muons, respectively, on 7 TeV protons [4].

The enhanced high-mass photon luminosity at LHμC increases the production rates relative to the LHeC, with a channel-dependent enhancement. For example, the quoted elastic W^+W^- cross-sections are approximately 4.5 and 7 times larger for the 500 GeV and 1 TeV muon options, respectively, than for the 50 GeV LHeC benchmark. At equal integrated luminosity, these enhancements translate into larger produced samples; reconstructed yields also depend on acceptance and selection efficiencies. Beyond $\tau^+\tau^-$ and W^+W^-, the programme includes loop-induced $\gamma\gamma \to ZZ$, top-pair production $\gamma\gamma \to t\bar{t}$, and pair production of charged BSM states, including higgsinos and sleptons [4, 5]. The high-mass tails (as shown in Figure 4) motivate studies of tau electromagnetic properties and gauge-boson interactions and our anomalous-coupling deformations illustrate possible shape sensitivity.

5. BSM perspectives

The muon beam also provides access to interactions involving muons and proton constituents. Examples in HELP4CERN include resonant squark production through an R-parity-violating μq coupling and color-octet muon production through a μg interaction [5, 7]. These mechanisms illustrate the complementarity between single production at a muon-hadron collider and searches at proton colliders. Their quantitative reach depends on the model couplings, compositeness scale where applicable, luminosity and event selection.

6. Conclusions

The LHμC benchmarks combine an enlarged DIS phase space with substantial Higgs and top production rates. The dedicated $h \to b\bar{b}$ analysis and exploratory combined fit demonstrate why Higgs precision is a central motivation, while also showing the importance of external width/coupling information and explicit fit assumptions. Electroweak, top and photon-induced measurements broaden the programme, with muon-coupled BSM channels supplying additional discovery opportunities. The present projections, however, motivate further work on detector response, backgrounds, systematic uncertainties and consistent analysis benchmarks before a complete experimental sensitivity programme can be established.

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